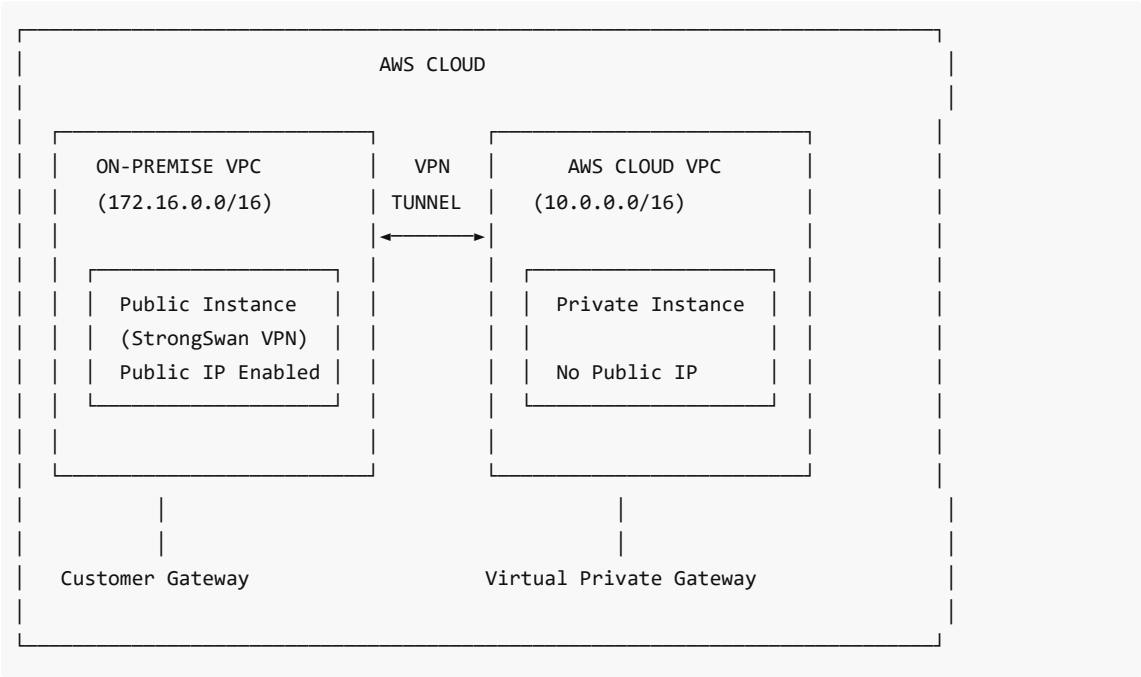


AWS Site-to-Site VPN Connection Setup Guide

Overview

This guide walks you through creating a **Site-to-Site VPN connection** between an On-Premise network (simulated in AWS) and an AWS Cloud VPC.

Architecture Diagram



Key Components

Component	Purpose
On-Premise VPC	Simulates your on-premise data center
AWS Cloud VPC	Your main cloud infrastructure
Customer Gateway	Represents your on-premise VPN device
Virtual Private Gateway	AWS-managed VPN endpoint
Site-to-Site VPN	Encrypted tunnel between both networks
StrongSwan	Open-source IPsec VPN software

Prerequisites

- AWS Account with appropriate permissions
- Basic understanding of VPC, Subnets, and Security Groups
- SSH key pair for EC2 instances

Step 1: Create On-Premise VPC

This VPC simulates your on-premise data center network.

1. Go to **VPC Console** → Click **Create VPC**
2. Select **VPC and more**
3. Configure:

Setting	Value
VPC Name	On-Premise-VPC
IPv4 CIDR Block	172.16.0.0/16
Availability Zones	1
Public Subnets	1
Private Subnets	0
NAT Gateways	None
VPC Endpoints	None

4. Click **Create VPC**
-

Step 2: Create AWS Cloud VPC

This is your main AWS cloud infrastructure.

1. Go to **VPC Console** → Click **Create VPC**
2. Select **VPC and more**
3. Configure:

Setting	Value
VPC Name	AWS-Cloud-VPC
IPv4 CIDR Block	10.0.0.0/16
Availability Zones	1
Public Subnets	0
Private Subnets	1
NAT Gateways	None
VPC Endpoints	None

4. Click **Create VPC**
-

Step 3: Create Public Instance (On-Premise - VPN Server)

This instance will act as the VPN gateway using StrongSwan.

1. Go to **EC2 Console** → Click **Launch Instance**

2. Configure:

Setting	Value
Name	On-Premise-Public-Instance
AMI	Amazon Linux 2 or Ubuntu
Instance Type	t2.micro
Key Pair	Create new or select existing
VPC	On-Premise-VPC
Subnet	Public Subnet
Auto-assign Public IP	Enable

3. **Security Group Configuration** - Create **Public-SG** :

Type	Protocol	Port Range	Source	Description
SSH	TCP	22	0.0.0.0/0	SSH Access
Custom UDP	UDP	500	0.0.0.0/0	IKE (IPsec)
Custom UDP	UDP	4500	0.0.0.0/0	NAT-T (IPsec)
All ICMP - IPv4	ICMP	All	10.0.0.0/16	Ping from AWS VPC

4. Click **Launch Instance**

5. **Note down the Public IP address** (e.g., **X.X.X.X**)

Step 4: Create Private Instance (AWS Cloud)

This instance is in the private subnet and will communicate through VPN.

1. Go to **EC2 Console** → Click **Launch Instance**

2. Configure:

Setting	Value
Name	AWS-Cloud-Private-Instance
AMI	Amazon Linux 2 or Ubuntu
Instance Type	t2.micro
Key Pair	Create new or select existing
VPC	AWS-Cloud-VPC
Subnet	Private Subnet
Auto-assign Public IP	Disable

3. Security Group Configuration - Create Private-SG :

Type	Protocol	Port Range	Source	Description
SSH	TCP	22	172.16.0.0/16	SSH from On-Premise
All ICMP - IPv4	ICMP	All	172.16.0.0/16	Ping from On-Premise

4. Click **Launch Instance**

Step 5: Create Customer Gateway

The Customer Gateway represents your on-premise VPN device (the public instance).

1. Go to **VPC Console** → **Customer Gateways** → **Create Customer Gateway**
2. Configure:

Setting	Value
Name	On-Premise-CGW
BGP ASN	65000 (default)
IP Address	Public IP of On-Premise Instance
Device	Leave empty

3. Click **Create Customer Gateway**

Step 6: Create Virtual Private Gateway

The Virtual Private Gateway is the AWS-managed VPN endpoint.

1. Go to **VPC Console** → **Virtual Private Gateways** → **Create Virtual Private Gateway**
2. Configure:

Setting	Value
Name	AWS-VPG
Amazon default ASN	Use default

3. Click **Create Virtual Private Gateway**

4. **Attach to VPC:**

- Select the VPG → **Actions** → **Attach to VPC**
 - Select **AWS-Cloud-VPC**
 - Click **Attach**
-

Step 7: Create Site-to-Site VPN Connection

1. Go to **VPC Console** → **Site-to-Site VPN Connections** → **Create VPN Connection**
2. Configure:

Setting	Value
Name	My-VPN-Connection
Target Gateway Type	Virtual Private Gateway
Virtual Private Gateway	AWS-VPG
Customer Gateway	Existing
Customer Gateway ID	On-Premise-CGW
Routing Options	Static
Static IP Prefixes	172.16.0.0/16

3. Click **Create VPN Connection**
4. Wait for the VPN to be created (takes 2-3 minutes)

Step 8: Download VPN Configuration

1. Select the VPN Connection → **Download Configuration**
2. Choose:

Setting	Value
Vendor	StrongSwan
Platform	Ubuntu
Software	StrongSwan
IKE Version	IKEv1

3. Download and save the configuration file

Step 9: Configure StrongSwan on Public Instance

9.1 Connect to Public Instance and Install StrongSwan

```
# SSH into your public instance
ssh -i your-key.pem ec2-user@<PUBLIC_IP>

# Update and install StrongSwan (Ubuntu)
sudo apt update && sudo apt install strongswan -y

# For Amazon Linux
sudo yum install strongswan -y
```

9.2 Enable IP Forwarding

```
# Edit sysctl.conf
sudo nano /etc/sysctl.conf

# Uncomment or add this line:
net.ipv4.ip_forward=1

# Apply changes
sudo sysctl -p

# Verify
cat /proc/sys/net/ipv4/ip_forward
# Should output: 1
```

9.3 Configure IPsec (from downloaded config)

Edit the IPsec configuration file:

```
sudo nano /etc/ipsec.conf
```

Add the tunnel configuration from downloaded file. Example structure:

```
conn Tunnel1
    auto=start
    left=%defaultroute
    leftid=<YOUR_PUBLIC_IP>
    right=<AWS_VPN_ENDPOINT_1>
    type=tunnel
    leftauth=psk
    rightauth=psk
    keyexchange=ikev1
    ike=aes128-sha1-modp1024
    ikelifetime=8h
    esp=aes128-sha1-modp1024
    lifetime=1h
    keyingtries=%forever
    leftsubnet=172.16.0.0/16
    rightsubnet=10.0.0.0/16
    dpddelay=10s
    dpdtimeout=30s
    dpdaction=restart
    leftupdown="/etc/ipsec.d/aws-updown.sh -ln Tunnel1 -ll <LOCAL_TUNNEL_IP>/30 -lr
<REMOTE_TUNNEL_IP>/30 -m 100 -r 10.0.0.0/16"
```

9.4 Configure IPsec Secrets

```
sudo nano /etc/ipsec.secrets
```

Add the pre-shared keys (from downloaded config):

```
# Tunnel 1
<YOUR_PUBLIC_IP> <AWS_ENDPOINT_1> : PSK "your-pre-shared-key-1"

# Tunnel 2
<YOUR_PUBLIC_IP> <AWS_ENDPOINT_2> : PSK "your-pre-shared-key-2"
```

9.5 Create AWS Updown Script

```
sudo nano /etc/ipsec.d/aws-updown.sh
```

Add the script content (from downloaded config) and make it executable:

```
sudo chmod 744 /etc/ipsec.d/aws-updown.sh
```

9.6 Start/Restart IPsec Service

```
# Restart IPsec
sudo ipsec restart

# Check status
sudo ipsec status

# View connection details
sudo ipsec statusall
```

Step 10: Update Route Table for Private Subnet

1. Go to **VPC Console** → **Route Tables**
2. Select the route table associated with `AWS-Cloud-VPC` private subnet
3. Click **Edit routes** → **Add route**

Destination	Target
172.16.0.0/16	Virtual Private Gateway (AWS-VPG)

4. Click **Save changes**
-

Step 11: Test VPN Connection

From Public Instance (On-Premise):

```
# Ping the private instance in AWS Cloud VPC
ping <PRIVATE_INSTANCE_PRIVATE_IP>
# Example: ping 10.0.1.50
```

From Private Instance (AWS Cloud):

```
# First, connect via Session Manager or bastion
# Then ping the on-premise public instance private IP
ping <PUBLIC_INSTANCE_PRIVATE_IP>
# Example: ping 172.16.1.100
```

Check VPN Status in AWS Console:

1. Go to **VPC Console** → **Site-to-Site VPN Connections**
2. Check **Tunnel Details** tab
3. Status should show **UP** for at least one tunnel

Troubleshooting

Issue	Solution
Tunnel shows DOWN	Check Security Group UDP ports 500 & 4500
Cannot ping across VPCs	Verify route tables have correct routes
IPsec service not starting	Check /var/log/syslog for errors
Connection timeout	Verify Customer Gateway IP is correct

Useful Commands:

```
# Check IPsec status
sudo ipsec status

# View detailed logs
sudo tail -f /var/log/syslog | grep -i ipsec

# Restart IPsec service
sudo ipsec restart

# Enable IP forwarding temporarily
sudo su
echo 1 > /proc/sys/net/ipv4/ip_forward
```

Cleanup / Delete Resources

Important: Delete resources in the correct order to avoid dependency errors.

Deletion Order:

Step	Resource	Action
1	EC2 Instances	Terminate both instances
2	Site-to-Site VPN Connection	Delete the VPN connection

3	Virtual Private Gateway	Detach from VPC, then delete
4	Customer Gateway	Delete the customer gateway
5	VPCs	Delete both VPCs (subnets, route tables, IGW will be deleted automatically)

Detailed Cleanup Steps:

1. Terminate EC2 Instances:

- EC2 Console → Instances → Select instances → Actions → Terminate

2. Delete Site-to-Site VPN Connection:

- VPC Console → Site-to-Site VPN Connections → Delete

3. Delete Virtual Private Gateway:

- VPC Console → Virtual Private Gateways → Detach from VPC → Delete

4. Delete Customer Gateway:

- VPC Console → Customer Gateways → Delete

5. Delete VPCs:

- VPC Console → Your VPCs → Delete VPC (deletes associated resources)

Summary

Step	Action
1	Create On-Premise VPC (172.16.0.0/16)
2	Create AWS Cloud VPC (10.0.0.0/16)
3	Create Public Instance in On-Premise VPC
4	Create Private Instance in AWS Cloud VPC
5	Create Customer Gateway (on-premise VPN device)
6	Create Virtual Private Gateway (AWS VPN endpoint)
7	Create Site-to-Site VPN Connection
8	Download StrongSwan configuration
9	Configure StrongSwan on public instance
10	Update route tables
11	Test connectivity

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